

# PROGRAMMING IN HASKELL

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Topic 4.4 - Structuring a solution on addresses  
using First Class Functions



# Example - structuring a solution

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Problem: We want to write an address generating function.

This function is to take names and addresses from the three different SETU campuses:

- Waterford Campus ,
- Carlow Campus ,
- Wexford Campus

with slightly different rules for different colleges.

We wish to structure a solution that generalises as much as possible, but also allows for differences.

We will pass functions into other functions to help!



## Example - structuring a solution

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### SETU (Waterford Campus):

Staff are allocated offices based (only) on their last name, i.e.

- < "L" – they are in the "Lower Floor, Main Building",
- otherwise, they are in the "Middle Floor, Science Building".

The remaining part of the address is

"SETU, Cork Road, Waterford, Ireland, X91 K0EK."



## Example - structuring a solution

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e.g. "Adam Able" from SETU (Waterford) would be given the address :

"Adam Able , Lower Floor, Main Building, SETU, Cork Road, Waterford, Ireland, X91 K0EK."

e.g. "Sean Stack" from SETU (Waterford) would be given the address :

"Sean Stack , Middle Floor, Science Building, SETU, Cork Road, Waterford, Ireland, X91 K0EK."



## Example - structuring a solution

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### **SETU (Carlow Campus):**

All staff SETU (Carlow Campus) have the same address:

"SETU (Carlow Campus), Dublin Road, Carlow, Ireland, R93 V960."



## Example - structuring a solution

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### SETU (Wexford Campus):

All staff in the Wexford Campus have the same address:

"SETU (Wexford Campus), Summerhill Rd, Townparks, Wexford, Y35 KA07."

In the address, however, the last name is before the first name.

e.g. "Adam Able" from SETU (Wexford Campus) would be given the address :

"Able, Adam, SETU (Wexford Campus), Summerhill Rd, Townparks, Wexford, Y35 KA07."



## Example - structuring a solution - A few notes:

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We use the (“Tom”, “Hanks”) structure to model a name and use a type synonym Name to encapsulate this structure.

```
type Name = (String, String)
```

When we write a function, we write the arguments in the order (left to right) from most general to least general. This allows maximum reuse of these functions.



## Example - structuring a solution

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Each location needs a different address structure. To calculate the full (String) address we need

- location
- name (not so obvious but in case of SETU ( Waterford Campus) , the address depends also on name)

We will use the following 'key's to indicate the locations:

- "WatSETU",
- "CarSETU" and
- "WexSETU"

(depending on the key, we will call a different function)



# Example - structuring a solution

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Top-down approach:

```
addressLetter :: Name -> String -> String  
addressLetter name location = getLocation location name
```

e.g. of running addressLetter:

```
ghci>  
ghci> addressLetter ("Siobhan", "Drohan") "Wat(SETU)"  
"Siobhan Drohan, Lower Floor, Main Building SETU (Waterford Campus), Cork Road, Waterford, Ireland, X91 K0EK."  
ghci>  
ghci> addressLetter ("Siobhan", "Roche") "Wat(SETU)"  
"Siobhan Roche, Top Floor, Main Building SETU (Waterford Campus), Cork Road, Waterford, Ireland, X91 K0EK."  
ghci>  
ghci> addressLetter ("Catherine", "Moloney") "Car(SETU)"  
"Catherine Moloney, SETU (Carlow Campus), Dublin Road, Carlow, Ireland, R93 V960."  
ghci>  
ghci> addressLetter ("Veronica", "Campbell") "Wex(SETU)"  
"Campbell, Veronica, SETU (Wexford Campus), Summerhill Rd, Townparks, Wexford, Y35 KA07."  
ghci> █
```



# Example - structuring a solution

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```
getLocation :: String -> (Name -> String)
getLocation location = case location of
  "Wat(SETU)" -> watOffice
  "Car(SETU)" -> carOffice
  "Wex(SETU)" -> wexOffice
  _           -> (\name -> fst name ++ " " ++ snd name ++ ": Address unknown" )
```

Returns the appropriate String producing function

Uses lambda function, expecting another argument



## Example - structuring a solution

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```
watOffice :: Name -> String
watOffice name =
  if lastName < "L"
    then nametext ++ ", Lower Floor, Main Building " ++ office WatSETU
    else nametext ++ ", Top Floor, Main Building " ++ office WatSETU
  where
    nametext = fst name ++ " " ++ snd name
    lastName = snd name
```



## Example - structuring a solution

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```
carOffice :: Name -> String
carOffice name =  nametext ++ ", " ++ office CarSETU
  where
    nametext = fst name ++ " " ++ snd name
```

```
wexOffice :: Name -> String
wexOffice name =  nametext ++ ", " ++ office WexSETU
  where
    nametext = snd name ++ ", " ++ fst name
```



## Example - structuring a solution

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```
data Location = WatSETU | CarSETU| WexSETU
:
:
office :: Location -> String
office WatSETU = "SETU (Waterford Campus), Cork Road, Waterford, Ireland, X91 K0EK."
office CarSETU= "SETU (Carlow Campus), Dublin Road, Carlow, Ireland, R93 V960."
office WexSETU = "SETU (Wexford Campus), Summerhill Rd, Townparks, Wexford, Y35 KA07."
```



## Example - Overall solution

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The full text of the solution is available in the associated lab



